EERoom

Browser-Based Electrical Engineering Simulation Platform

Final Project Proposal

Author	Aiden Vargas
Course Number	CST-452 Senior Project II
Instructor	Professor Thompson
Revision	2.0 (final, supersedes 1.0 from Milestone 1)
Date	November 4, 2026
Live Application	https://www.eeroom.io
Source Repository	https://github.com/AidenVargas03/EERoom

Grand Canyon University | College of Engineering and Technology

Abstract

EERoom is a free, browser-based electrical engineering simulation platform. It gives students three interactive tools that run in any modern browser with nothing to install: an Ohm's Law and power calculator, a waveform visualizer that demonstrates aliasing, and a logic gate sandbox that builds a truth table and Boolean expression as the circuit is wired. Anyone can use the tools as a guest. A free account adds saving, a dashboard of saved projects, password recovery, and read-only share links.

This final revision of the proposal records the project as delivered. The original proposal planned four tools, Google sign-in, and a paid Pro tier through Stripe. After the first code iteration the scope was narrowed to the three tools and core platform features that one developer could finish, test, and deploy well; the reasons are recorded in Project Scope and in the Requirements Specification. The application is live at www.eeroom.io, is covered by 119 automated tests and 27 manual test cases, and is documented in a public GitHub repository and a project portfolio page.

History and Signoff Sheet

Change Record

Date	Author	Revision Notes
CST-451	Aiden Vargas	1.0 Initial proposal (Milestone 1)
11/04/2026	Aiden Vargas	2.0 Final proposal updated to the delivered project (Milestone 6)

Overall Instructor Feedback and Comments

Integrated instructor feedback into project documentation	☐ Yes ☐ No
Project approval	☐ Professor Thompson
Submitted by	Aiden Vargas, 11/04/2026

Contents

Project Overview and Project Objectives
Project Scope
Project Success Measures
Project High-Level Solution
Project Controls
Project Cost and Schedule
Appendix A: References
Appendix B: Copyright Compliance

Project Overview and Project Objectives

Problem and Background

Engineering students meet Ohm's Law, sampling, and digital logic early, mostly as equations. Industry simulation tools such as MATLAB, Ansys, and Cadence are licensed per seat, installed on lab computers, and slow to learn, so many students have no simple way to experiment on their own. Free alternatives are often dated or narrow.

EERoom fills that gap with focused tools that open in a browser on any computer or tablet. A student can change a value and see the result immediately, save the work, and send a read-only link to a classmate or instructor.

Christian Worldview

EERoom reflects the principle of stewardship and service. Removing cost and installation as barriers gives every student the same chance to practice, whatever their means or school. Proverbs 11:14 notes that "in an abundance of counselors there is safety" (English Standard Version, 2001), and EERoom's share links let students learn from one another's work.

The project also reflects Colossians 3:23, "Whatever you do, work heartily, as for the Lord and not for men" (English Standard Version, 2001). That principle shaped how the scope was handled: rather than delivering more features of uncertain quality, the project was narrowed so that what was delivered is tested, documented, secure, and honest about its limits. Protecting users' data, refusing to reveal who has an account, and keeping the tools free all treat users with care.

Project Objectives and Outcome

#	Original Objective (Revision 1.0)	Outcome
1	Browser circuit simulator with DC analysis	Removed (SC-01): highest risk, could not be finished and tested in time
2	Ohm's Law and power calculator	Delivered
3	Logic gate sandbox with truth tables	Delivered, with 7 gates, Boolean expressions, and loop and floating-pin checks
4	Waveform visualizer on an oscilloscope display	Delivered, with sample rate and aliasing demonstration
5	Google OAuth and email login	Email and password delivered; Google OAuth removed (SC-02)
6	Cloud saving and shareable URLs	Delivered, plus guest access with save on sign-up
7	Live deployment at a custom domain	Delivered at www.eeroom.io
8	Freemium model with Stripe	Removed (SC-03); every feature is free
9	Public GitHub repository with documentation	Delivered, with README and project portfolio page

Challenges

- Scope: the original plan was too large for one developer. It was narrowed after the first iteration and the changes were recorded and justified.

- Deployment: configuring CORS, environment variables, and email redirects across Vercel, Railway, and Supabase took several rounds of fixes.
- Security details: rate limiting behind a proxy, ownership checks on every query, and a reset flow that does not reveal accounts.
- Solo development: all design, coding, testing, deployment, and documentation were done by one person.

Benefits and Opportunities

- Free, installation-free practice for students who lack licensed software.
- A portfolio project that demonstrates full-stack development, automated testing, deployment, and security practice to employers.
- An extensible design: each tool stores its state as JSON in one table, so new tools, or rooms for other disciplines, can be added without database changes.

Project Scope

Final Scope Statement

EERoom Version 1.0 is a browser-based electrical engineering platform with three simulation tools, email and password accounts with password recovery, guest access, cloud project saving with a dashboard, and read-only share links, deployed at a custom domain and documented in a public repository. All development was done by one developer with free and open-source technologies.

In Scope (Delivered)

- Ohm's Law and Power Calculator, Waveform Visualizer, and Logic Gate Sandbox
- Guest use of every tool, with the guest's work saved after sign-up
- Email and password registration, login, logout, and password reset
- Saving, reopening, renaming, and deleting projects from a dashboard
- Read-only share links that work without an account
- Layouts that work on desktop and tablet (768 px and wider), with WCAG 2.2 AA text contrast
- Deployment on Vercel, Railway, and Supabase at www.eeroom.io
- Public GitHub repository, README, and project portfolio page

Removed From Scope

Item	Reason
DC circuit simulator (SC-01)	Highest-risk feature; nodal analysis for any user-built circuit could not be finished and tested in time
Google OAuth (SC-02)	Email and password login meets the requirement
Stripe, Pro tier, save limit, PDF export (SC-03)	Not needed to show the platform works; removes a payment security risk

Out of Scope

- Native mobile apps
- Phone layouts below 768 px
- Tools for other engineering disciplines
- AC or SPICE-level circuit analysis
- Real-time multi-user collaboration
- LMS integration

Stakeholders

Stakeholder	Role	Responsibilities
Aiden Vargas	Student, developer, project manager	All design, development, testing, deployment, and documentation

Stakeholder	Role	Responsibilities
Professor Thompson	Instructor	Milestone review, feedback, approval, and grading
Capstone mentor	Mentor	Review and approval of scope changes
Engineering students	End users	Use the tools for coursework and give feedback at the Technology Showcase

Work Breakdown Structure (Final Status)

ID	Task	Status	Completed In
1	Milestone 1: Project Proposal	Complete	CST-451
2	Milestone 2: Requirements Specification	Complete	CST-451
3	Milestone 3: Architecture Plan	Complete	CST-451
4	React app and Express API setup	Complete	CST-451, iteration 1
5	Circuit simulator	Removed (SC-01)	N/A
6	Ohm's Law calculator	Complete	CST-451, iteration 1
7	Logic gate sandbox	Complete	Milestone 4, iteration 2
8	Waveform visualizer	Complete	Milestone 4, iteration 2
9	Authentication and database	Complete	Iteration 1; reset in iteration 3
10	Project save, dashboard, and share	Complete	Milestone 4, iteration 2
11	Stripe payments	Removed (SC-03)	N/A
12	Guest access, tablet layout, API tests	Complete	Milestone 5, iteration 3
13	Deployment and custom domain	Complete	Milestone 4
14	Final documentation, portfolio, presentation	Complete	Milestone 6

Hours for CST-452 were tracked in the Sprint Backlog: 113 hours estimated and 99 hours recorded across Milestones 4 and 5.

Project Success Measures

Original Completion Criterion	Final Result
All four simulation tools work at the live URL	Met for the three tools in the final scope; the circuit simulator was removed (SC-01)
Sign up, login, logout, and password reset work	Met (MT-01 to MT-05, MT-18 to MT-21; IT-01 to IT-15)
Users can create, save, load, and delete projects	Met (MT-08 to MT-10; IT-21 to IT-31)
Share URLs load the correct project for any user	Met (MT-11 to MT-13; IT-32 to IT-37)
Stripe gates Pro features	Removed with the Pro tier (SC-03)
Deployed and publicly accessible at a custom domain	Met: www.eeroom.io
Responsive on desktop and mobile	Met for desktop and tablet (MT-26, MT-27); phones below 768 px are out of scope
Milestones submitted on time	Met through Milestone 5; Milestone 6 is this submission
Public repository with code, README, and documentation	Met
Final demonstration of all tools and features	Covered in the final presentation

Assumptions and Constraints

#	Description	Type	Final Status
1	Free cloud tiers remain available for development	Assumption	Held for Vercel and Supabase; Railway's trial ends, so the API moves to its low-cost paid plan
2	Circuit simulation limited to DC	Assumption	Closed: circuit simulator removed
3	Continuous access to a development machine	Assumption	Held
4	Stripe test mode only	Assumption	Closed: Stripe removed
5	Work completed within the course schedule	Constraint	Met
6	Out-of-pocket budget under $300	Constraint	Met
7	Free and open-source tools only	Constraint	Met
8	One developer	Constraint	Met

Project High-Level Solution

Introduction

EERoom turns fundamentals usually taught on paper into tools a student can experiment with. All calculations run in the browser, so results are instant; the server stores and returns projects and handles accounts.

Architecture

- Presentation tier: React 18 single page application with Tailwind CSS, served by Vercel at www.eeroom.io. The logic board is drawn in SVG and the waveform display on an HTML5 canvas.
- Application tier: Node.js and Express REST API on Railway with public, rate-limited /auth routes, JWT-protected /projects routes, and a public read-only /share route.
- Data tier: PostgreSQL and Supabase Auth on Supabase. Each project's state is stored as JSON in one projects table.

The Final Architecture Plan gives the full design with diagrams, the schema, every API endpoint, and the security design.

Tools Delivered

- Ohm's Law and Power Calculator: enter any two of V, I, and R to get the third and the power, with guards against division by zero.
- Waveform Visualizer: sine, square, sawtooth, and triangle signals with amplitude, frequency, phase, and offset; a sample rate control overlays what that rate records and warns when the signal is aliased.
- Logic Gate Sandbox: AND, OR, NOT, NAND, NOR, XOR, and XNOR on a grid, with a live truth table, a Boolean expression per output, floating-pin warnings, and refusal of feedback loops.

Technology Stack

Layer	Technology	Cost
Frontend	React 18, React Router 6, Vite 5, Tailwind CSS 3	Free
Backend	Node.js, Express 4, jsonwebtoken, express-rate-limit	Free
Database and auth	PostgreSQL and Supabase Auth	Free tier
Testing	Vitest, Supertest	Free
Frontend hosting	Vercel	Free tier
Backend hosting	Railway	About $5 per month after the trial
Domain	eeroom.io	Annual registration
Version control	Git and GitHub (public)	Free

Project Controls

Risk Management (Final)

Risk	Probability / Impact	What Happened
Circuit simulation exceeds the time available	Medium / High	Occurred. Handled by removing the simulator (SC-01) and recording the decision
Third-party service outage	Low / Medium	No outages affected development; the API's /health route confirms the deployment
Stripe integration delays launch	Low / Medium	Avoided by removing payments (SC-03)
Scope creep	High / High	Controlled: changes were recorded in the scope tables and submitted for approval
Loss of work	Low / High	Avoided: every change was committed and pushed to GitHub
Hosting trial ends before grading	Medium / High	Mitigation: move Railway to its paid plan before grading so the site stays live

Issues Log

#	Issue	Resolution	Status
1	Linux build failed on a filename case mismatch in App.jsx	Fixed the import case	Resolved
2	Registration could leave an orphaned auth user if the users row failed	Added a rollback that deletes the auth user	Resolved
3	Deep links and share URLs returned 404 on Vercel	Added an SPA rewrite in vercel.json	Resolved
4	Rate limiter saw every user as the proxy's address	Set trust proxy to 1 for Railway	Resolved
5	Reset email linked to the default localhost address	Set PASSWORD_RESET_URL on Railway and the Site and Redirect URLs in Supabase	Resolved
6	Logic board cut off on tablets	Board scales to fit; pin targets kept at 24 px	Resolved

Change Control Log

ID	Change	Decision	Included In
SC-01	Remove the circuit simulator	Submitted for approval at Milestone 4	Requirements Rev 1.1
SC-02	Remove Google OAuth	Submitted for approval at Milestone 4	Requirements Rev 1.1

ID	Change	Decision	Included In
SC-03	Remove Stripe, Pro tier, save limit, PDF export	Submitted for approval at Milestone 4	Requirements Rev 1.1
SC-04	SVG grid instead of react-konva	Submitted for approval at Milestone 4	Requirements Rev 1.1
SC-05	Ideal-versus-sampled overlay	Submitted for approval at Milestone 4	Requirements Rev 1.1
SC-06	Guest access deferred, then built	Resolved at Milestone 5	Requirements Rev 1.2
SC-07	Password reset form deferred, then built	Resolved at Milestone 5	Requirements Rev 1.2
SC-08	Vitest instead of Jest for API tests	Submitted at Milestone 5	Requirements Rev 1.2

Roles and Responsibilities

Name	Role	Responsibility
Aiden Vargas	Developer and project manager	All design, frontend, backend, testing, deployment, and documentation
Professor Thompson	Instructor	Milestone review, feedback, approval, and grading
Capstone mentor	Mentor	Scope change approval

Project Cost and Schedule

Cost

Item	Vendor	Cost
Custom domain (eeroom.io)	Domain registrar	Annual registration
Backend hosting	Railway	Free trial, then about $5 per month
Frontend hosting	Vercel	Free tier
Database and authentication	Supabase	Free tier
Payment processing	Stripe	None (removed)

All costs stayed within the $300 budget constraint.

Schedule (Final)

Milestone	Key Deliverables	Status
Milestone 1: Project Proposal	Proposal, scope, schedule	Complete (CST-451)
Milestone 2: Requirements	Functional and non-functional requirements, user stories	Complete (CST-451)
Milestone 3: Architecture Plan	Design specification and first code iteration	Complete (CST-451)
Milestone 4: Coding and Testing	Second iteration: all tools, saving, sharing, deployment, scope review	Complete
Milestone 5: Coding and Testing	Third iteration: password reset, guest access, tablets, API tests	Complete
Technology Showcase Poster	Poster for the showcase	Complete
Milestone 6: Final Project and Presentation	Final documents, README, portfolio, presentation	Complete

Appendix A: References

Crossway. (2001). The Holy Bible, English Standard Version. Crossway Bibles.

Grand Canyon University. (2026). CST-452 capstone project handbook.

MathWorks. (2026). MATLAB. https://www.mathworks.com/products/matlab.html

Ansys, Inc. (2026). Ansys simulation software. https://www.ansys.com

Supabase. (2026). Supabase documentation. https://supabase.com/docs

Vercel. (2026). Vercel documentation. https://vercel.com/docs

Railway. (2026). Railway documentation. https://docs.railway.com

Appendix B: Copyright Compliance

All third-party libraries used in the final application are open source under permissive licenses that allow academic and commercial use. Code adapted from documentation is cited in the source files and listed in the README.

Library or Tool	License	How Used
React and React DOM	MIT	User interface
React Router	MIT	Page routing
Vite	MIT	Development server and build
Tailwind CSS	MIT	Styling
Axios	MIT	HTTP requests from the client
Express	MIT	REST API
jsonwebtoken	MIT	Signing and verifying login tokens
express-rate-limit	MIT	Rate limiting the auth routes
cors, dotenv	MIT, BSD-2-Clause	Allowed origins; environment variables
@supabase/supabase-js	MIT	Database and authentication client
Vitest, Supertest	MIT	Automated testing
IBM Plex Sans and Mono	SIL Open Font License	Fonts